Speed is an important advantage of in-place deployments with CodeDeploy. However, applying changes to long-running virtual machines, also called snowflake servers, is risky. A deployment might fail on a machine because of a change applied weeks before. It is difficult to reproduce the exact state of a machine.

That's why we prefer a different approach. To roll out a new revision, instead of modifying a running machine, spin up a new virtual machine. By doing so, each deployment starts from the same state, which increases reliability. You will learn how to implement a rolling update of EC2 instances with the help of CloudFormation in the following section.

Blue-green deployments with CodeDeploy

We have omitted that CodeDeploy supports blue-green deployments. With this method, a new machine is started instead of an in-place update. However, we prefer using CloudFormation for rolling updates because of its simplicity.

15.2 Rolling update with AWS CloudFormation and user data

CloudFormation is the Infrastructure as Code tool designed to manage AWS resources in an automated way. In addition, you can also use CloudFormation to orchestrate a rolling update of EC2 instances in an Auto Scaling group.